Adaptive Planar Search:

A Terminal-Placement Acquisition for Derivative-Free Optimization

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Abstract

Derivative-free optimizers spend most of their budget on one-dimensional line searches. We use the terminal-placement rule of the companion theory paper as a line-search inner loop: with its scale and correlation length estimated online, it flees, probes, or commits in one to three evaluations. On the HumpDay physics benchmark it is a competitive inner loop on rough and high-dimensional objectives at tight budgets, where its early exits buy more search directions than a classical line search. A projection argument then lifts the same rule off the line: after two observations under any radial Gaussian-process prior, the entire next-point decision lives in a plane, in any ambient dimension. A fixed off-line placement does not beat a line search across the catalog, but choosing it by expected improvement under a fitted Gaussian process does, and increasingly so with dimension: the value of the acquired point, not its position, is what turns the exact reduction into a win.

1 Line searches are where the budget goes

Derivative-free optimizers reduce a high-dimensional problem to a sequence of one-dimensional subproblems: pick a direction, place a few evaluations along it, return a point. Classical inner loops — golden section, Brent — assume smooth unimodal slices and spend eight to ten evaluations resolving each one. On the rough slices of real physics objectives that assumption fails, and on a tight global budget the cost of each line is what binds.

The companion theory paper solves a different one-dimensional problem exactly: place three evaluations on an exponentiated Ornstein–Uhlenbeck slice and be paid the value at the final point. Its policy flees a below-median start, probes between two good values, and stays above a ceiling, all in closed form. Read as an inner loop it costs one to three evaluations per line, not eight to ten. This paper asks what that buys in an optimizer, and how far off the line the same geometry reaches.

Two findings. As an inner loop the rule is competitive on rough and high-dimensional objectives when the budget is tight, for a reason that is about evaluation economy more than about the placement theorem. And the placement geometry is not confined to a line: after two observations under a radial prior the next-point decision collapses to a plane, exactly, in any ambient dimension. A fixed off-line placement does not pay, but one aimed by the information content of the acquired point — a fitted Gaussian process — beats a line search on three quarters of high-dimensional tasks, which is where the reduction should matter.

Objective	d	`grass3`	rank	best rival	seed wins
plinko_funnel	7	-57.0	1/5	golden6 -50.8	13/16
wind_farm	16	-94.1	1/5	golden6 -93.6	16/24
bowling	4	-105.0	1/5	golden2 -104.0	4/8
free_kick	4	-32.9	5/5	golden2 -105.2	3/24

Table 1: `grass3` as an inner loop, median objective over seeds at budget 120; lower is better. Comparators are golden section (two and six evaluations) and Brent in the shared outer loop, and whole-cube random search. Rank is among the five, averaging tied medians; seed wins count seeds on which `grass3` beats the best rival. Across all 22 objectives `grass3` has mean rank 2.95/5 and is first or second on eight (nine counting joint second places); `free_kick` is a representative loss.

2 The terminal-placement inner loop

We implement the three-evaluation rule as an inner loop, `grass3`: the exact final placement and opening step of the theory paper, with the mean, scale, and correlation length estimated online from the optimizer's own history. It flees or stays after one or two evaluations when the rule dictates and places the third otherwise. It is a heuristic adaptation, not the exact policy: it interpolates the opening-correlation table, extrapolates it above the tabulated range, forces flight after three stalled lines, and clips placements to the unit cube. The last is a genuine approximation — choosing an unconstrained exterior winner and then clipping can lose to the best feasible interior point — and we flag it rather than hide it.

We benchmark `grass3` inside a common outer loop, iterated random-direction search at budget 120, against four fixed inner loops: golden section with two and with six evaluations, Brent, and random placement, on matched directions and seeds.

3 Benchmark

The objectives are the application suite of the HumpDay package (Cotton 2021): small physics and engineering problems in which a point in the unit cube sets a deterministic simulator's controls and the objective is the outcome — a break in pool, pins felled in bowling, a trebuchet's range, a wind-farm layout. We use the 22 dynamics-simulating problems; their one-dimensional slices run from smooth to collision-cascade rough.

The rule wins on two kinds of objective. On collision and agent-chaos objectives with rough but structured slices it is the best inner loop; on a plinko objective it leads its strongest rival on 13 of 16 seeds (Table 1). It also wins on high-dimensional objectives even when slices are smooth: on a sixteen-dimensional wind-farm layout it leads golden6 on 16 of 24 seeds, and golden2 and Brent on 21 and 22.

The high-dimensional win is evaluation economy. Early exits of one or two evaluations let the outer loop try several times more directions than an inner loop spending eight to ten per line. The win is budget-starved rather than general: it appears when the dimension times a classical line's cost exceeds the whole budget, so a method with a fixed number of directions per iteration cannot complete one pass.

The rule loses in two regimes. On landscapes too rough to carry navigable structure, broad sampling wins and placement theory has nothing to exploit; on needle landscapes, where the value hides in a spike, golden section's fixed full-segment probes are the right move. It is also not a drop-in for the line search inside an existing optimizer: substituted into Powell, whose direction updates need genuine line minimization, it loses to Brent.

Three qualifications keep the reading honest. Random search samples the whole cube rather than sharing the outer loop, and the Brent baseline spends one call reevaluating the known incumbent, so the economy margin is somewhat overstated. An ablation is more sobering: omitting the third placement altogether helps on twelve of the twenty-two objectives, sometimes by a wide margin, so the benchmark speaks for evaluation economy and online adaptation more than for the placement theorem in particular. A control on exact exponentiated-OU fields, the model's own landscape, loses to broad probing — terminal-placement wins do not come from the objectives resembling an OU path.

As a standalone optimizer rather than an inner loop, a budget-truncated two-shot form is mid-field against a panel of eleven standard methods, best at the smallest budgets and overtaken by population methods as the budget grows — the expected place for a rule built from a few-evaluation terminal-placement model.

4 From lines to planes

The line is not special. Fix a Gaussian-process prior with constant mean and radial covariance $K(\|x-y\|)$, and observe values at $x_1, \ldots, x_n$. Let H be their affine hull and write a candidate as $x = h + w$ with $h \in H$ and $w \perp H$. Then

$$\|x - x_i\|^2 = \|h - x_i\|^2 + \|w\|^2, \tag{1}$$

so the candidate's posterior depends only on its position within H and its distance from H. Every perpendicular direction is equivalent.

Proposition 1 (Dimension reduction). Under a radial Gaussian-process prior on an unrestricted domain, a single-point acquisition can be optimized in at most n dimensions after n observations, regardless of the ambient dimension.

With two observations the next-point decision lives in a plane, in any ambient dimension, and needs no OU structure. The companion theory paper computes that planar decision exactly for the exponential kernel: after observing values b, d at correlation ρ, the candidate's correlations (x, y) to the two observations range over the triangle inequalities $\{xy \leq \rho,\ x \geq \rho y,\ y \geq \rho x\}$, whose boundary is the one-dimensional bracket and two exterior rays and whose interior holds one extra candidate, the point that matches both observed values.

That extra candidate is worth a measurable amount. With $b = d = \rho = \frac{1}{2}$ the optimal third point forms an equilateral triangle with the two observations, and its conditional expected terminal reward exceeds the best collinear choice by about 2.9%. The gain is a planar effect: it is unavailable to any line search.

5 An adaptive planar search

The reduction suggests a move. Evaluate a trial along a chosen direction; use its observed value to choose the next point within a plane containing that line; under the isotropic model, take any perpendicular direction. The plane costs one extra coordinate and, by the dimension-reduction proposition, captures the whole next-point decision no matter how large the ambient space.

One adjustment is necessary, and the opening constant closes. The theory paper rewards the terminal recommendation, whereas an optimizer keeps the best evaluated point, so its opening maximizes $\mathbb{E}[\max(e^b, e^X)]$ over the probe correlation ρ, with $X \sim N(b\rho, 1-\rho^2)$. The geometric reduction is unaffected — it is a fact about the posterior, not the objective — but the opening coefficient is not.

Proposition 2 (Expected-improvement opening). For the best-evaluated objective the optimal opening correlation satisfies $\rho^*(b) = \kappa_{\mathrm{EI}}\, b + O(b^3)$ as $b \downarrow 0$, with

$$\kappa_{\mathrm{EI}} = \frac{\mathbb{E}[e^Z; Z>0]}{\mathbb{E}[Z\, e^Z; Z>0]} = \frac{\Phi(1)}{\Phi(1)+\phi(1)} = 0.776639\ldots, \qquad Z \sim N(0,1). \tag{2}$$

The first-order condition $\partial_\rho \mathbb{E}[\max(e^b, e^X)] = \mathbb{E}\Big[e^X \mathbf{1}\{X > b\}\,(b - \rho X'/\sqrt{1-\rho^2})\Big] = 0$, with $X = b\rho + \sqrt{1-\rho^2}X'$ and $X' \sim N(0,1)$, gives at $b \downarrow 0$ the ratio $\mathbb{E}[e^Z; Z>0] = \kappa_{\mathrm{EI}}\,\mathbb{E}[Ze^Z; Z>0]$, and $\mathbb{E}[e^Z; Z>0] = e^{1/2}\Phi(1)$, $\mathbb{E}[Ze^Z; Z>0] = e^{1/2}(\phi(1)+\Phi(1))$. So the optimizer opens *wider* than the terminal rule — $\kappa_{\mathrm{EI}} = 0.777$ against the theory paper's $\kappa_0 = 0.540$ — because a fresh high draw is worth more when the best point is kept rather than a mean recommendation. And it is elementary, where κ_0 is only characterized by a transcendental equation. A full planar-search implementation is the remaining empirical step.

6 Does the off-line placement help?

A budget-matched experiment isolates the planar move: two evaluations per step along a shared random direction, differing only in where the second goes. The *line* keeps it on the line, extrapolating past the trial when it improved and reflecting when it did not; the *plane* puts it at a fixed equilateral apex off the line. Start point, direction stream, and adaptive step are identical. Across the whole catalog — 51 analytic objectives at $d \in \{2, 5, 10\}$ and 69 physics simulators at native dimensions, 222 tasks — the fixed off-line move wins on only 0.26 of seeds, below one half at every dimension (0.30 at $d \le 3$, 0.21 at $d \ge 8$), and a momentum-estimated transverse direction is no better.

The fixed move ignores information. The reduction says the decision lives in the plane, but a fixed apex in a random plane sends the one off-line probe where the objective's structure almost never is; the reduction is a fact about the posterior, not about any particular objective.

7 Aiming the plane with information

Choose the off-line point by information instead of geometry. Fit a Gaussian process to the evaluated history and, among candidate points in the plane, take the one of highest expected improvement — the value-of-information criterion behind the entropy-search family (Hennig and Schuler 2012; Wang and Jegelka 2017). The control is the same GP restricted to the line: identical guidance and budget, with only the off-line candidates removed, so the difference isolates the value of the off-line freedom.

dimension	off-line beats on-line
$d = 2$–3	0.42
$d = 4$–7	0.63
$d \geq 8$	0.76
overall	0.62

Table 2: Fraction of seeds on which the GP-aimed off-line placement beats a GP restricted to the line, over the 222-task catalog at ten seeds each. Both use the same GP guidance and budget; only the off-line candidates differ. The off-line freedom helps more as the dimension grows, on analytic objectives (0.78 at $d \geq 8$) and simulators (0.71) alike.

Information inverts the fixed-geometry result (Table 2). Off-line placement is neutral in two dimensions, where a line-restricted GP already covers the small space, and it wins on three quarters of seeds by eight dimensions and above, against a control that guides the on-line point just as well. The geometry hands high-dimensional search a plane; information decides where in it to look; and the payoff grows in exactly the regime where a line search has the least room. What matters is the value of the acquired point, not its position, which is the reading the entropy-search literature would predict, and here it is what turns an exact reduction into a win.

8 Related work

LineBO (Kirschner et al. 2019) optimizes through adaptively selected one-dimensional subspaces; the reduction here identifies when one extra transverse coordinate carries measurable value, sharpening the choice of subspace to a plane after two observations. The Stochastic Three Points method (Bergou et al. 2020) compares an incumbent with two opposite trials; the planar candidate suggests an adaptive triangle variant whose third location is chosen after seeing the second value. Knowledge gradient (Frazier, Powell and Dayanik 2009) supplies the measure-then-choose framework; the contribution here is explicit geometry and the resulting planar move within it.

9 Scope

The defensible claim for high-dimensional derivative-free optimization is an exact simplification of the next decision under a radial prior: whatever the ambient dimension, two observations reduce it to a plane. Learning useful directions over a long run remains hard, and constraints or unknown anisotropy break the isotropy the reduction uses. The immediate application is therefore small evaluation budgets and local searches, where a single well-placed plane is most of what a step can afford.

References

[1] P. Cotton. When the grass is greener: three-shot search on exponentiated Gaussian landscapes. Working paper, 2026. https://github.com/microprediction/browniansearch.

[2] P. Cotton. HumpDay: pure Python derivative-free optimization and its application suite. https://github.com/microprediction/humpday, 2021.

[3] J. Kirschner, M. Mutný, N. Hiller, R. Ischebeck, and A. Krause. Adaptive and safe Bayesian optimization in high dimensions via one-dimensional subspaces. In *Proceedings of the 36th International Conference on Machine Learning*, 2019.

[4] E. H. Bergou, E. Gorbunov, and P. Richtárik. Stochastic three points method for unconstrained smooth minimization. *SIAM Journal on Optimization*, 30(4):2726–2749, 2020.

[5] P. I. Frazier, W. B. Powell, and S. Dayanik. The knowledge-gradient policy for correlated normal beliefs. *INFORMS Journal on Computing*, 21(4):599–613, 2009.

[6] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[7] P. Hennig and C. J. Schuler. Entropy search for information-efficient global optimization. *Journal of Machine Learning Research*, 13:1809–1837, 2012.

[8] Z. Wang and S. Jegelka. Max-value entropy search for efficient Bayesian optimization. In *Proceedings of the 34th International Conference on Machine Learning*, 2017.